Paper 1: No calculator

4. [Maximum mark: 5]

In the expansion of $(x+k)^7$, where $k \in \mathbb{R}$, the coefficient of the term in x^5 is 63.

Find the possible values of k.

Paper 2: Use a calculator

6. [Maximum mark: 5]

Consider the expansion of $(3+x^2)^{n+1}$, where $n \in \mathbb{Z}^+$.

Given that the coefficient of x^4 is $20\,412$, find the value of n.

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4.13 Classwork: **Binomial distribution**

3. a. A box holds 240 eggs. The probability that an egg is brown is 0.05.

Find the expected number of brown eggs in the box. *[2 marks]*

b. Find the probability that there are 15 brown eggs in the box. *[2 marks]*

c. Find the probability that there are at least 10 brown eggs in the box. *[3 marks]*

4a. The following table shows the probability distribution of a discrete random variable *X*.

x	0	2	5	9
$\mathrm{P}(X = x)$	0.3	k	$2k$	0.1

Find the value of *k*. *[3 marks]*

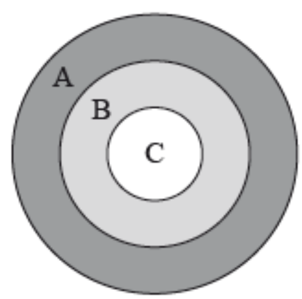
b. Find $\mathrm{E}(X)$. *[3 marks]*

5. The random variable X has the following probability distribution.

x	1	2	3
$\mathrm{P}(X = x)$	s	0.3	q

Given that $\mathrm{E}(X) = 1.7$, find *q*. *[6 marks]*

6a. The following diagram shows a board which is divided into three regions A, B and C.



A game consists of a contestant throwing one dart at the board. The probability of hitting each region is given in the following table.

Region	A	B	C
Probability	$\frac{5}{20}$	$\frac{4}{20}$	$\frac{1}{20}$

Find the probability that the dart does **not** hit the board. *[3 marks]*

b. The contestant scores points as shown in the following table.

Region	A	B	C	Does not hit the board
Points	0	q	10	−3

Given that the game is fair, find the value of *q*. *[4 marks]*

Paper 1: No calculator

2. [Maximum mark: 4]

Consider two consecutive positive integers, n and $n + 1$.

Show that the difference of their squares is equal to the sum of the two integers.